POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Physics

Semester: Spring

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why is really S.H.M rare? Deduce the expression for the time period of a compound pendulum and radius of gyration of compound pendulum. 9
   b) A boy claps his hands once every second and hears the echo from a distant building. He hears the echo of each clap mid-way between it and the next clap. If the velocity of sound is 340m/s, what is the distance of the building from the body? 6
2. a) What is diffraction? Differentiate between Fresnel and Fraunhofer diffraction pattern. Discuss the intensity distribution in the diffraction pattern due to single slit. 9
   b) Newton's rings that are formed by sodium light between the flat glass plate and convex lens are viewed normally from above. What will be the order of the dark ring, which will have double the diameter of 40th dark ring? 6
3. a) Discuss the types of optical fiber and explain the working principle of optical fiber. 9
   b) A hall has a volume $2265m^3$. Its total absorption is equivalent to $92.9m^2$ of open window. What will be the effects of reverberation time if and audience fills the hall and thereby increase the absorption by another $92.9m^2$? 6
4. a) Show that the magnitude of electric potential along axial line due to linear quadrupole is double than the potential along equatorial line. 9
   b) Obtain the charging time constant of a capacitor in a RC circuit such that current through the resistor is decreased by half of its peak value in 5 seconds. 6
5. a) State and explain Biot-Savart's law. Use it to find the expression for magnetic field on the axis of current carrying coil. 9
   b) Find the strength of magnetic field at the center of rectangular coil of length l and width d, which carries current I. 6

6. a) Write the Maxwell's equation in differential form and their significance. Using Maxwell's equations prove that $\frac{E_m}{B_m} = c$ Where symbol carry usual meaning. 9
   b) Discuss the Sabine's formula to obtain the expression for growth and decay of sound energy in room. 6
7. Write short notes on: **(Any two)** 2×5
   a) Classify solids on the basis of band theory of solids
   b) Tunneling effect
   c) Lorentz force

Testpage